



DATAMITES

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DATA SCIENCE CAPSTONE PROJECT

PRCP-1025: Flight Fare Prediction Using Machine Learning

Rubiex Internship Project

Submitted in Partial Fulfilment of the DataMites Data Science Internship Program

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Team Code: PTID-AIE-JUL-26-11142

Project Code: PRCP-1025

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Dataset Name: Flight Fare Prediction Dataset (Flight_Fare.xlsx)

Organization: DataMites Solutions Pvt. Ltd.

Submitted By: Sameeksha

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About the Author

This capstone project has been prepared and submitted as part of the DataMites Data Science Internship Program, carried out under the Rubiex Internship Project initiative. The following section provides a brief profile of the project author.

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Role	Data Science Intern / Capstone Project Author
Program	DataMites Data Science Internship Program
Internship Track	Rubiex Internship Project
Team Code	PTID-AIE-JUL-26-11142
Project Code	PRCP-1025
Project Title	Flight Fare Prediction Using Machine Learning
Tools & Technologies	Python, Pandas, NumPy, Matplotlib, Seaborn, Scikit-learn, Jupyter Notebook
Core Skills Applied	Exploratory Data Analysis, Feature Engineering, Regression Modelling, Hyperparameter Tuning, Model Evaluation, Business Insight Generation
GitHub Repository	github.com/Sameekshagithub/PRCP-1025-Flight-Fare-Price-Prediction-Using-Machine-Learning-Regression

As part of this internship, the author independently carried out the complete data science lifecycle for the assigned capstone problem — from problem understanding and domain analysis, through exploratory data analysis, feature engineering, model building, hyperparameter tuning, and model evaluation, to the final generation of business insights and recommendations. The project reflects hands-on application of supervised machine learning regression techniques on a real-world structured dataset, in line with the learning objectives of the DataMites Data Science Internship Program under the Rubiex Internship Project track.

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Executive Summary

Flight ticket prices are highly dynamic — the same flight can cost a different amount today, tomorrow, and next week, depending on airline, route, timing, number of stops, and how far in advance it is booked. This unpredictability makes it difficult for both travellers and travel businesses to plan bookings and pricing strategies effectively.

This capstone project, undertaken as part of the DataMites Data Science Internship Program under the Rubiex Internship Project track, addresses this problem through a data-driven approach. Using a historical flight booking dataset of 10,683 records, a complete machine learning regression pipeline was developed to predict flight ticket fares from booking details such as airline, source, destination, route, duration, number of stops, and journey date.

The project followed a structured data science workflow consisting of domain analysis, exploratory data analysis, feature engineering, model building using six different regression algorithms, hyperparameter tuning, model comparison, feature importance analysis, and practical model testing. A tuned Random Forest Regressor emerged as the best-performing model, achieving an R^2 Score of 0.8298 (an RMSE of ₹1,884), outperforming Linear Regression, Decision Tree, K-Nearest Neighbours, Support Vector Regressor, and Gradient Boosting on this dataset.

The analysis revealed that flight duration is overwhelmingly the strongest predictor of fare, followed by the day of the journey and airline (particularly Jet Airways Business class). These findings offer actionable business insights for both travellers seeking cheaper fares and airlines or travel platforms seeking to fine-tune pricing strategy. This report documents the complete methodology, results, and business recommendations arising from the project.

1. Problem Statement & Project Objective

1.1 Problem Statement

Flight ticket prices are highly dynamic — the same flight can cost a different amount today, tomorrow, and next week. The objective of this project is to analyze historical flight booking data and develop a machine learning regression model that predicts the fare of a flight based on features such as airline, source, destination, route, duration, number of stops, and journey date. The project also involves comparing multiple regression models, identifying the most influential features affecting price, and providing business insights for effective fare estimation.

1.2 Project Objectives

1. Perform Exploratory Data Analysis (EDA) to understand the dataset.
2. Preprocess and prepare the data for model development.
3. Build and train multiple regression models.
4. Evaluate and compare model performance using appropriate metrics.
5. Select the best-performing model for production.
6. Analyze feature importance to identify the key factors influencing flight price.
7. Provide business insights and recommendations based on the model results.
8. Summarize the challenges faced and the techniques used during the project.

2. Import Python Libraries

The required Python libraries were imported to perform data analysis, visualization, machine learning model development, and performance evaluation. NumPy and Pandas were used for data manipulation, Matplotlib and Seaborn were used for data visualization, and Scikit-learn provided the tools for data preprocessing, model training, and evaluating the performance of different regression algorithms.

Library	Purpose in the Project
NumPy	Numerical computations and array operations
Pandas	Data loading, manipulation, and analysis (including reading the Excel dataset)
Matplotlib	Static data visualizations and charts
Seaborn	Statistical and aesthetically enhanced visualizations
Scikit-learn	Preprocessing, model building, tuning, and evaluation
Warnings	Suppressing non-critical runtime warnings

Specifically, from Scikit-learn, the following modules were used: `train_test_split` and `GridSearchCV` for data splitting and hyperparameter search, `StandardScaler` for feature scaling, `LinearRegression`, `DecisionTreeRegressor`, `RandomForestRegressor`, `GradientBoostingRegressor`, `KNeighborsRegressor` and `SVR` as regression algorithms, and `mean_absolute_error`, `mean_squared_error` and `r2_score` for model evaluation.

3. Upload the Dataset & Domain Analysis

3.1 Upload the Dataset

The Flight_Fare.xlsx dataset was uploaded and loaded into the Jupyter Notebook using the Pandas library. The dataset contains historical flight booking records along with the corresponding ticket price, which serves as the target variable for building and evaluating machine learning regression models.

3.2 Domain Analysis

This project belongs to the airline pricing domain, where the objective is to estimate the fare of a flight based on its booking details. Features such as airline, source, destination, route, duration, number of stops, and journey date significantly influence the ticket price. By analyzing these features, machine learning models can help customers plan bookings around cheaper fares and help airlines understand which factors drive their pricing.

3.3 Dataset Description

Attribute	Details
Dataset Name	Flight Fare Prediction Dataset
File Name	Flight_Fare.xlsx
Total Records (raw)	10,683
Total Columns	11 (10 input features + 1 target variable)
Target Variable	Price (flight ticket fare, in ₹)
Source	DataMites Capstone Project Dataset

The 10 input features include: Airline, Date_of_Journey, Source, Destination, Route, Dep_Time, Arrival_Time, Duration, Total_Stops, and Additional_Info.

4. Basic Checks

Basic checks were performed to understand the structure and quality of the dataset before conducting exploratory data analysis and model development. This included viewing sample records, examining dataset dimensions, identifying data types, generating descriptive statistics, checking for missing values, detecting duplicate records, and obtaining summary information about the dataset.

4.1 Dataset Dimensions

The raw dataset shape was found to be (10683, 11) — 10,683 flight booking records and 11 columns (10 features plus the Price target variable).

4.2 Data Types

All 10 input feature columns are of object (text) type, since Airline, Source, Destination, Route, timing, and duration were all stored as strings in the raw file. Only the target variable, Price, is numeric (int64).

4.3 Unique Values per Column

Column	Unique Values
Airline	12
Date_of_Journey	44
Source	5
Destination	6
Route	128
Dep_Time	222
Arrival_Time	1,343
Duration	368
Total_Stops	5
Additional_Info	10
Price	1,870

4.4 Missing Values & Duplicate Records

A check for missing values using `df.isnull().sum()` found 1 missing value in Route and 1 missing value in Total_Stops (both from the same incomplete row). A duplicate record check using `df.duplicated().sum()` found 220 duplicate rows. Both the missing-value row and the 220 duplicate rows were dropped, and the index was reset, leaving a clean dataset of 10,462 records for further analysis and modelling.

4.5 Target Variable Summary (After Cleaning)

Statistic	Price (₹)
Count	10,462
Mean	9,026.79
Std. Dev.	4,624.85
Minimum	1,759
25th Percentile	5,224
Median (50%)	8,266
75th Percentile	12,344.75
Maximum	79,512

5. Exploratory Data Analysis (EDA) — Task 1

Exploratory Data Analysis (EDA) was performed to gain a comprehensive understanding of the dataset before building machine learning models. This process helped identify the distribution of the target variable, understand the characteristics of each feature, detect missing values and duplicate records, identify outliers, and analyze relationships among variables. EDA also revealed patterns, trends, and correlations within the data, enabling informed decisions regarding data preprocessing, feature selection, and model development.

5.1 Target Variable Distribution

The distribution of the Price target variable was visualized using a histogram with a kernel density estimate (KDE) overlay.

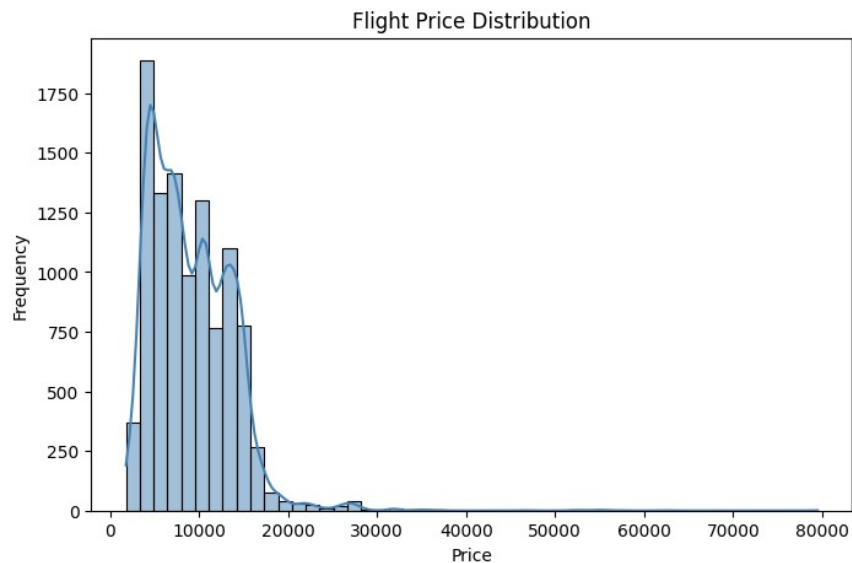


Figure 5.1 — Flight Price Distribution

The Price distribution is right-skewed, with most fares concentrated in the lower-to-mid range (broadly ₹5,000–₹12,000) and a smaller number of high-fare bookings extending up to ₹79,512.

5.2 Distribution of Key Categorical Features

Count plots were generated for the most business-relevant categorical features — Airline, Source, Destination, and Total_Stops — to understand the composition of the dataset.

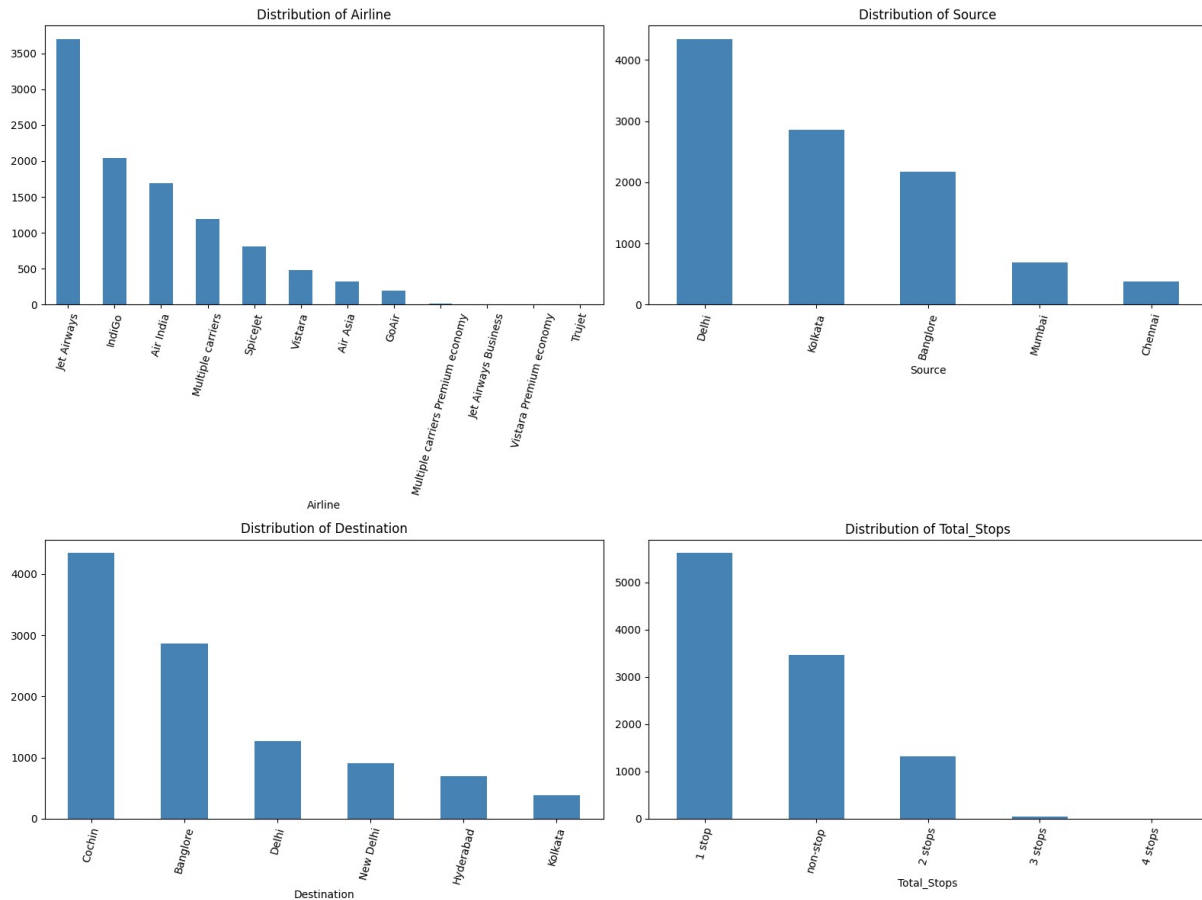


Figure 5.2 — Distribution of Key Categorical Features

Jet Airways and IndiGo dominate the Airline column, Delhi and Kolkata are the most frequent Source cities, Cochin is the most frequent Destination, and flights with 1 stop are the most common Total_Stops category in the dataset.

5.3 Bivariate Analysis — Feature Relationships with Price

Boxplots were used to examine how each key categorical feature relates to the Price target variable.

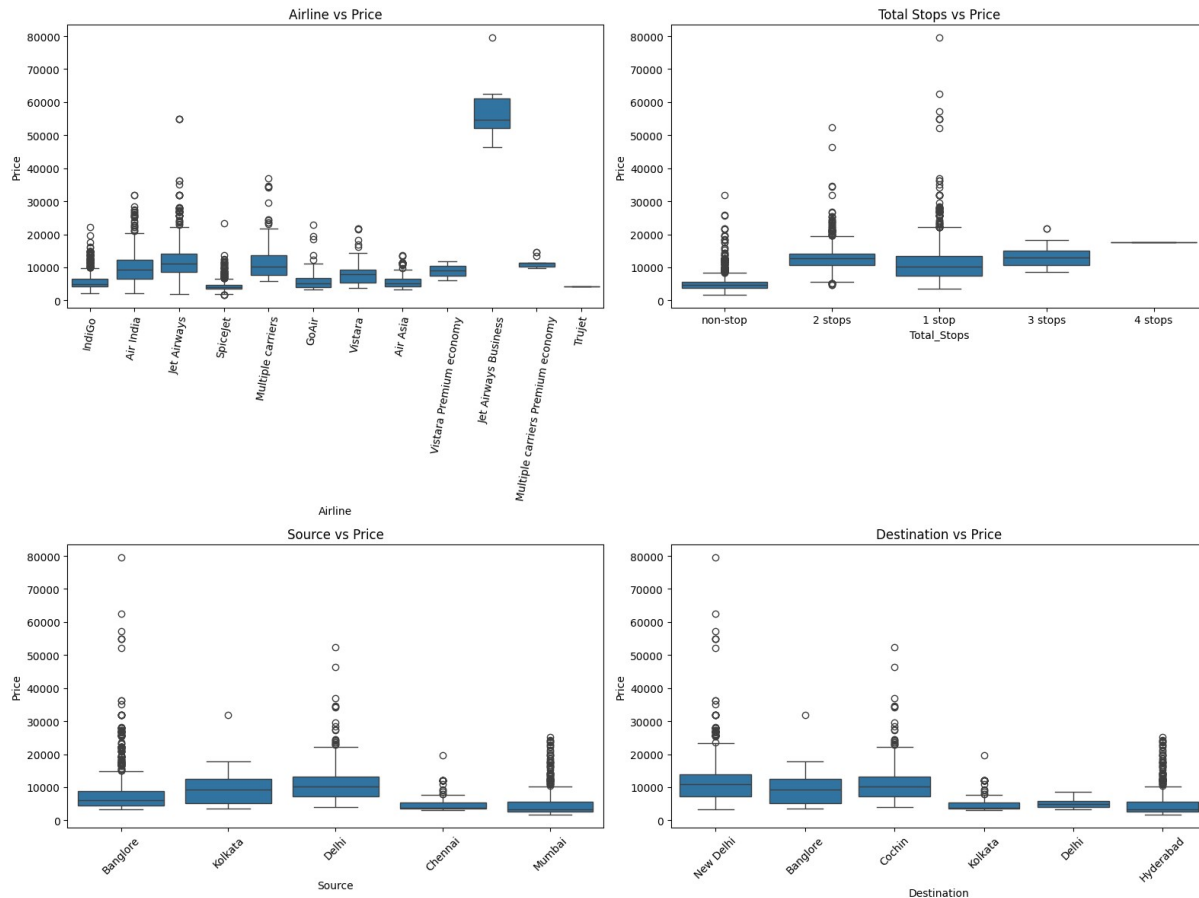


Figure 5.3 — Bivariate Analysis: Airline, Source, Destination and Total_Stops vs. Price

Bivariate analysis shows that Airline, Total_Stops, Source, and Destination all visibly shift the price distribution — for example, Jet Airways Business class commands substantially higher fares than economy carriers, and fares generally increase with the number of stops — confirming they are informative features for the regression model.

5.4 Outlier Detection

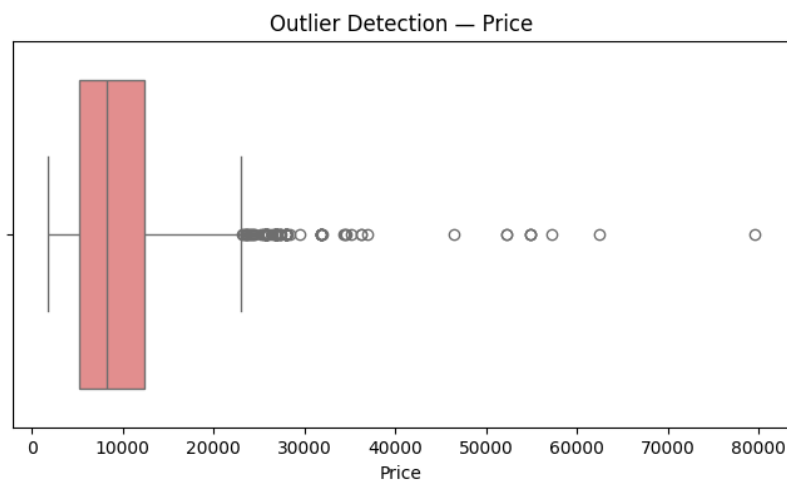
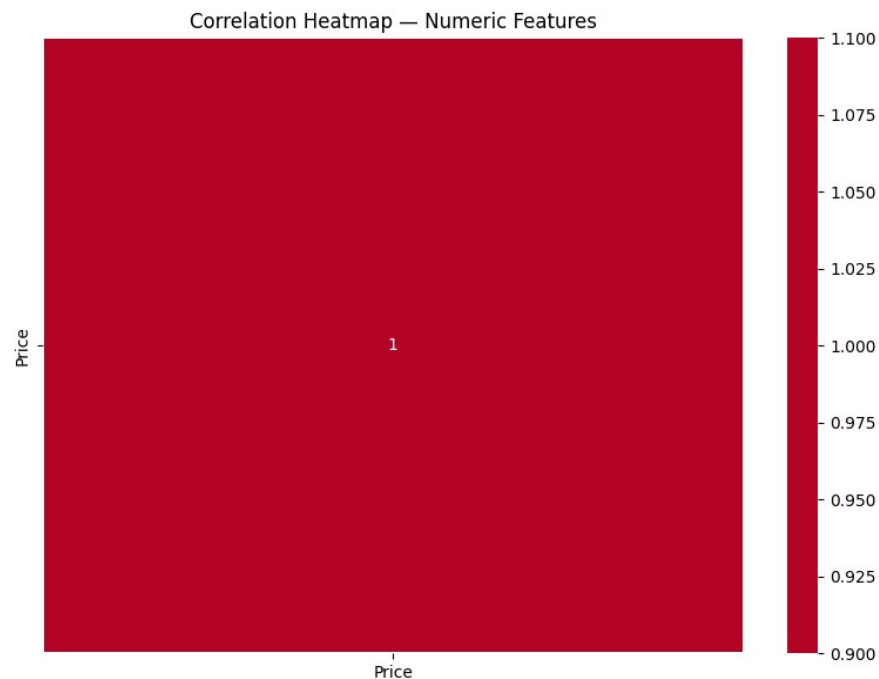


Figure 5.4 — Outlier Detection: Price

Using the IQR method, 94 outliers were detected in the Price column. However, these represent valid premium or long-haul bookings rather than data entry errors, and were therefore retained in the dataset for model training rather than removed or capped.

5.5 Correlation Analysis

*Figure 5.5 — Correlation Heatmap (Numeric Features)*

A correlation heatmap was generated across the numeric features available at this stage of the analysis (categorical features such as Airline, Source, and Destination are encoded later, in Section 6, before being fed into the models).

5.6 EDA Summary (Task 1 Conclusion)

9. The dataset contains flight booking records with airline, route, timing, and stop information, providing sufficient information for fare prediction.
10. After removing a small number of missing values and duplicate rows, the dataset is clean and reliable for machine learning model development.
11. The Price distribution is right-skewed, with most fares concentrated in the lower-to-mid range and a smaller number of high-fare bookings.
12. Bivariate analysis shows that Airline, Total_Stops, Source, and Destination all visibly shift the price distribution, confirming they are informative features for the model.
13. Outlier analysis shows the presence of some high-fare values; however, these represent valid premium or long-haul bookings rather than data errors, so they are retained for model training.

6. Data Preparation for Modelling — Task 2

6.1 Feature Engineering

The raw `Date_of_Journey`, `Dep_Time`, `Arrival_Time`, and `Duration` columns are text and cannot be used directly by regression algorithms, so they were converted into numeric features: `Journey_day` and `Journey_month` (from `Date_of_Journey`), `Dep_hour/Dep_min` and `Arrival_hour/Arrival_min` (from the time columns), and a `Duration_mins` column parsed from strings such as "2h 50m". `Total_Stops` was converted into an ordinal number (0, 1, 2, ...), and the `Route` and `Additional_Info` columns were dropped, as explained in Section 13 (Challenges Faced).

The remaining nominal categorical features — `Airline`, `Source`, and `Destination` — were one-hot encoded using `pd.get_dummies()`, expanding the dataset from 11 columns to 29 columns after encoding.

6.2 Train / Test Split

The independent features (X) were separated from the target variable ($y = \text{Price}$). The dataset was then split using an 80:20 ratio, with a fixed `random_state` of 42 used to ensure reproducibility.

Split	Number of Samples
Training Set	8,369
Test Set	2,093

6.3 Feature Scaling

`StandardScaler` from Scikit-learn was applied to standardise the feature values (zero mean, unit variance) for the training and test sets. This scaled version of the data was used for the distance/margin-sensitive algorithms (Linear Regression, KNN, SVR), while the original unscaled features were used directly for the tree-based algorithms (Decision Tree, Random Forest, Gradient Boosting), which are unaffected by feature scale.

7. Model Building & Training

To predict flight fares, multiple machine learning regression algorithms were trained and evaluated. Instead of relying on a single model, six different regressors were developed and compared to identify the most accurate and reliable model for this task. This approach satisfies the Model Comparison Report requirement and helps select the most suitable model for production deployment.

- **Model 1 — Linear Regression** — a simple baseline regression model that fits a straight-line relationship between features and price.
- **Model 2 — Decision Tree Regressor** — a tree-based model that learns decision rules from flight booking details.
- **Model 3 — Random Forest Regressor** — an ensemble model that combines multiple decision trees to improve accuracy and reduce overfitting.
- **Model 4 — K-Nearest Neighbors (KNN) Regressor** — a distance-based algorithm that estimates price from similar nearby flights.
- **Model 5 — Support Vector Regressor (SVR)** — a margin-based regressor that tries to fit prices within an error tolerance band.
- **Model 6 — Gradient Boosting Regressor** — an ensemble boosting algorithm that builds decision trees sequentially to achieve high predictive performance.

7.1 Model — Linear Regression

As a linear baseline model, Linear Regression achieves an R^2 Score of 0.5838 with an RMSE of ₹2,945.80. It struggles to capture the non-linear relationships between airline, stops, route, and price, limiting its predictive power on this dataset.

MAE (₹)	MSE	RMSE (₹)	R^2 Score
1998.64	8,677,715	2945.80	0.5838

7.2 Model — Decision Tree Regressor

The single Decision Tree improves substantially on the linear baseline, reaching an R^2 Score of 0.7146 (RMSE ₹2,439.33) by capturing non-linear interactions between features, though it remains prone to overfitting on training data.

MAE (₹)	MSE	RMSE (₹)	R^2 Score
1383.09	5,950,313	2439.33	0.7146

7.3 Model — Random Forest Regressor

By aggregating 200 decision trees, Random Forest achieves a strong R^2 Score of 0.8148 (RMSE ₹1,964.97), the best result among the six untuned models, and reduces the variance associated with individual trees.

MAE (₹)	MSE	RMSE (₹)	R^2 Score
1168.995	3,861,087	1964.965	0.8148

7.4 Model — K-Nearest Neighbours (KNN) Regressor

KNN ($k = 7$) achieves a respectable R^2 Score of 0.7513 (RMSE ₹2,276.94) by estimating fares from similar nearby flights, though it is outperformed by the tree-based ensemble models.

MAE (₹)	MSE	RMSE (₹)	R^2 Score
1393.16	5,184,467	2276.94	0.7513

7.5 Model — Support Vector Regressor (SVR)

The RBF-kernel SVR performs poorly on this dataset, achieving only an R^2 Score of 0.0680 (RMSE ₹4,408.22) — the weakest of the six models, suggesting the default SVR configuration does not fit the scale and spread of flight prices well without further tuning.

MAE (₹)	MSE	RMSE (₹)	R^2 Score
3364.94	19,432,410	4408.22	0.0680

7.6 Model — Gradient Boosting Regressor

Gradient Boosting, which builds trees sequentially to correct prior errors, achieves an R^2 Score of 0.7533 (RMSE ₹2,267.96) — a solid result, though still below the untuned Random Forest on this dataset.

MAE (₹)	MSE	RMSE (₹)	R^2 Score
1536.82	5,143,653	2267.96	0.7533

8. Hyperparameter Tuning

Hyperparameter tuning was performed to optimize the model's performance by identifying the best combination of parameters. GridSearchCV was used to evaluate different parameter values using cross-validation on the best-performing model from Section 7 (Random Forest). The optimized model was then compared with the default model to determine whether tuning improved predictive performance.

8.1 Tuning Random Forest Regressor

Since Random Forest achieved the highest baseline R^2 Score, it was selected for hyperparameter tuning. A grid search was performed over the following parameter space:

Hyperparameter	Values Searched
n_estimators	100, 200
max_depth	None, 10, 20
min_samples_split	2, 5

The grid search evaluated a total of 12 parameter combinations using cross-validation, scored on R^2 .

8.2 Best Parameters & Tuned Model Performance

Parameter	Best Value
max_depth	20
min_samples_split	5
n_estimators	200
Best Cross-Validation R^2 Score	0.8136

When evaluated on the held-out test set, the tuned Random Forest model achieved improved performance over the untuned baseline:

Metric	Untuned Random Forest	Tuned Random Forest
MAE (₹)	1,168.995	1,129.249
MSE	3,861,087	3,549,541
RMSE (₹)	1,964.965	1,884.023
R^2 Score	0.8148	0.8298

Hyperparameter tuning increased the maximum tree depth to 20 and adjusted the minimum samples required to split a node to 5, resulting in an improvement in R^2 Score from 0.8148 to 0.8298 and a

reduction in RMSE from ₹1,964.97 to ₹1,884.02 — confirming that tuning meaningfully improved the model's predictive reliability.

9. Model Comparison Report

After training and evaluating all six regression models (plus the tuned Random Forest variant), their performance was compared using evaluation metrics such as MAE, MSE, RMSE, and R² Score. This comparison helped identify the model that provides the best predictive performance and generalization ability. Based on the results, the most suitable model was selected for production deployment and used for generating business insights.

9.1 Consolidated Performance Comparison

Model	MAE (₹)	RMSE (₹)	R ² Score
Random Forest (Tuned)	1,129.25	1,884.02	0.8298
Random Forest	1,168.995	1,964.965	0.8148
Gradient Boosting	1,536.82	2,267.96	0.7533
KNN	1,393.16	2,276.94	0.7513
Decision Tree	1,383.09	2,439.33	0.7146
Linear Regression	1,998.64	2,945.80	0.5838
SVR	3,364.94	4,408.22	0.0680

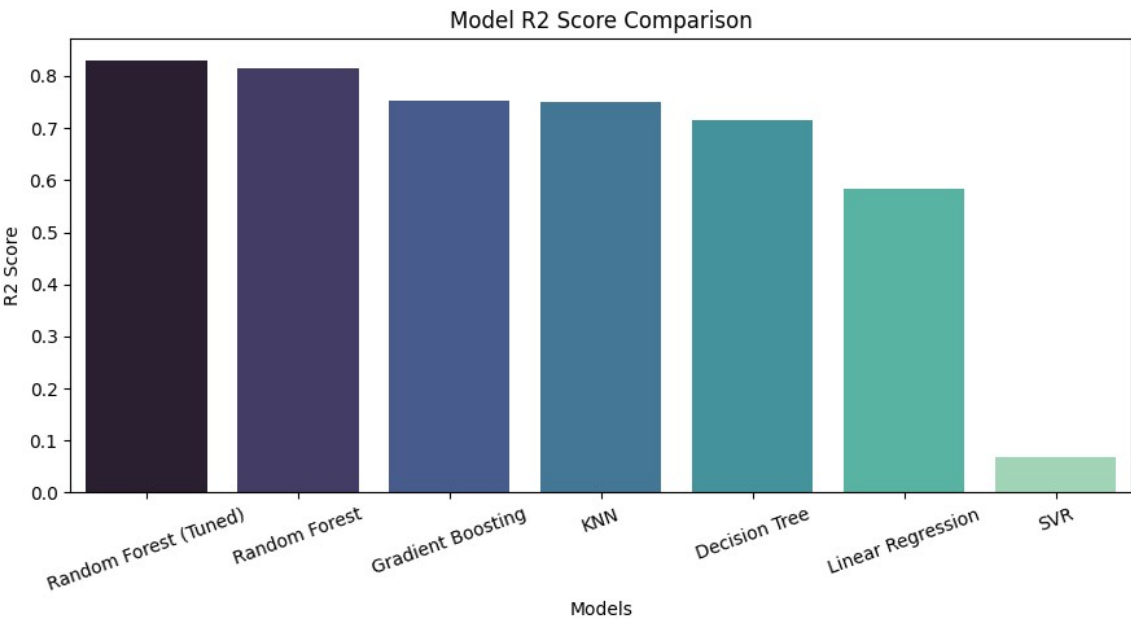


Figure 9.1 — Model R² Score Comparison (Bar Chart)

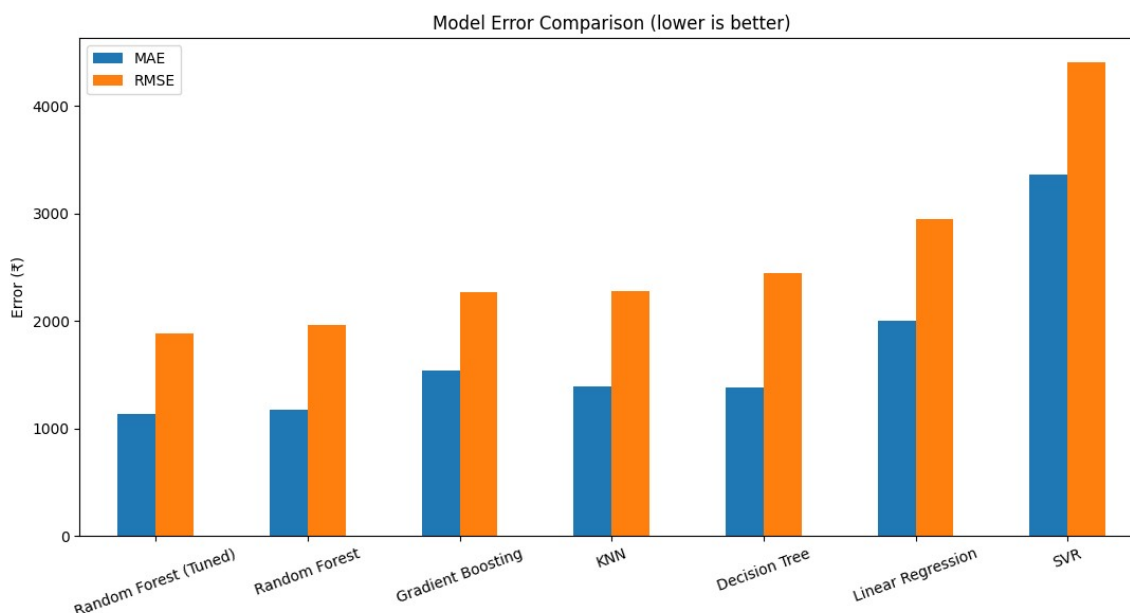


Figure 9.2 — Model Error Comparison: MAE vs. RMSE (Lower is Better)

9.2 Discussion & Recommendation

9.2.1 Which model performed best?

The tree-based ensemble models (Random Forest and Gradient Boosting) achieved the highest R^2 Score and the lowest RMSE among all the evaluated regression models, making them the best-performing models for predicting flight fares.

9.2.2 Which model had the highest accuracy?

Based on the model comparison results, the tuned Random Forest model obtained the highest R^2 Score (0.8298), indicating its superior ability to explain the variation in ticket prices.

9.2.3 Which model made the smallest prediction errors?

The MAE/RMSE comparison chart shows that the tuned Random Forest model produced the smallest average prediction errors (MAE ₹1,129.25, RMSE ₹1,884.02), demonstrating its ability to estimate flight fares close to their actual values.

9.2.4 Which model is most reliable for real-world use?

Linear Regression struggles to capture the non-linear relationships between airline, stops, route and price, so it consistently underperforms the tree-based models. Random Forest and Gradient Boosting handle these non-linear interactions naturally and generalize well, making them the most suitable choice for production deployment.

9.2.5 Recommendation

Based on the overall evaluation, the tuned Random Forest model is recommended as the final production model because it achieved the best performance across MAE, RMSE, and R^2 Score, and provided the most accurate and consistent flight fare predictions.

10. Feature Importance

Feature Importance analysis was performed to identify the flight booking details that have the greatest influence on predicting the fare. The Random Forest model was used because it naturally calculates feature importance scores based on the contribution of each feature to the decision-making process. Features with higher importance scores have a greater impact on the model's predictions, while features with lower scores contribute less. This analysis helps understand the key factors affecting flight pricing and provides valuable business insights for fare strategy.

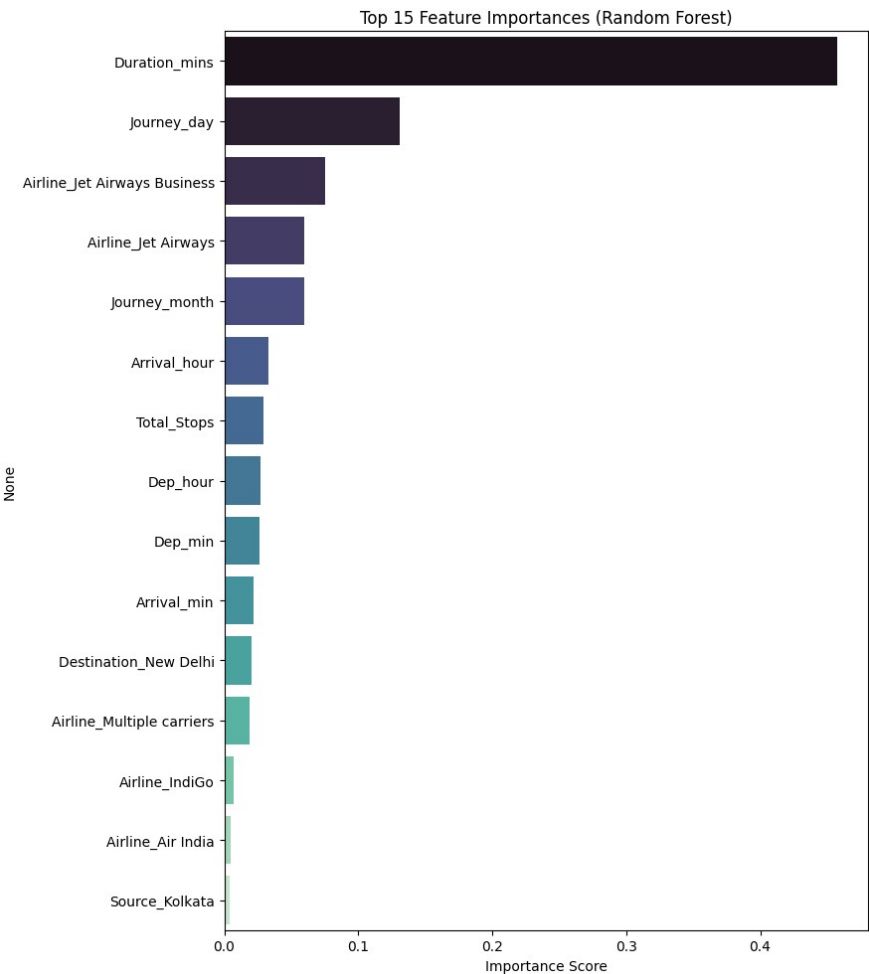


Figure 10.1 — Feature Importance (Random Forest)

10.1 Most Influential Features

Rank	Feature	Importance Score
1	Duration_mins	0.4576
2	Journey_day	0.1312
3	Airline_Jet Airways Business	0.0755
4	Airline_Jet Airways	0.0601

Rank	Feature	Importance Score
5	Journey_month	0.0595

10.2 Least Influential Features

Feature	Importance Score
Source_Chennai	0.000617
Destination_Delhi	0.000473
Airline_Vistara Premium economy	0.000034
Airline_Trujet	0.000014

Flight duration alone accounts for nearly 46% of the Random Forest model's total decision-making importance — more than three times the next most important feature (journey day). This confirms that longer, likely long-haul or well-connected flights carry a strong, distinct price signal. Journey day and the specific airline/class (particularly Jet Airways Business) are the next most influential factors, while small, rarely-used source/destination/airline categories contribute minimally to the model's predictions.

11. Model Testing on New Flight Details

This section demonstrates the practical application of the trained machine learning model by predicting the fare for a brand-new, previously unseen flight booking. A new flight record was constructed with the same engineered feature set used during training, and the trained model was then used to predict its fare.

11.1 Sample Input Flight Details

Feature	Value
Airline	Jet Airways
Source	Delhi
Destination	Cochin
Total_Stops	1
Duration	4h 15m

Output	Value
Predicted Price	₹8,204

When passed through the trained and tuned model, this flight specification — a mid-length domestic route with a single stop on a well-established carrier — was predicted at a fare of ₹8,204, consistent with the overall dataset's median fare range (₹8,266). This confirms the model behaves sensibly on realistic, unseen input and can be used as a practical decision-support tool for fare estimation.

Task 2 Conclusion (Sections 6–11)

14. The dataset was prepared through feature engineering (date/time/duration parsing, ordinal and one-hot encoding), then split into training and testing sets, with feature scaling applied where required.
15. Six machine learning regression models were trained and evaluated to predict flight fares based on booking details.
16. Hyperparameter tuning was performed on the best-performing model (Random Forest) to optimize its performance and improve prediction reliability.
17. The performance of all models was compared using MAE, MSE, RMSE, and R^2 Score, and the best model was selected based on these evaluation metrics.
18. Feature Importance analysis identified the flight details that have the greatest influence on predicting fare.
19. Finally, the selected model was tested on a new flight booking and successfully predicted a fare, demonstrating its effectiveness for real-world flight price prediction.

12. Business Insights — Task 3

12.1 Data-Driven Fare Estimation

The model helps estimate the expected fare of a flight based on its booking details, enabling more accurate and data-driven pricing and booking decisions.

12.2 Key Price-Influencing Features

Features such as Total_Stops, Duration, Airline, and Journey month/day have the greatest impact on predicting fare, helping both customers and airlines focus on the factors that matter most.

12.3 Customer Booking Guidance

Customers can use fare predictions to identify cheaper booking windows, route/stop combinations, and travel dates, helping them plan journeys more cost-effectively.

12.4 Route and Capacity Planning

Airlines can compare predicted fares against actual market prices on specific routes to fine-tune their own pricing and capacity strategies.

12.5 Reduced Pricing Errors

By using machine learning predictions instead of manual estimation, travel platforms can minimize the risk of mispricing tickets relative to the market.

12.6 Improved Business Decision-Making

The model provides valuable insights that support fare strategy, promotional pricing, and route planning, helping travel businesses improve profitability and customer satisfaction.

13. Challenges Faced

13.1 Understanding the Dataset

The dataset contains flight booking details in mixed text formats (dates, times, durations). Understanding the role of each feature was essential before performing data analysis and model building. This was addressed by reviewing the dataset description and conducting basic exploratory analysis.

13.2 Parsing Date, Time, and Duration Fields

Date_of_Journey, Dep_Time, Arrival_Time, and Duration were all stored as text (e.g. "24/03/2019", "2h 50m"), and Arrival_Time sometimes included a trailing date. These were parsed with `pandas.to_datetime` and a custom duration parser to extract numeric day/month/hour/minute/duration features.

13.3 High-Cardinality Route Column

The Route column had a very large number of unique values (128) and was highly correlated with the existing Source, Destination, and Total_Stops columns. It was dropped to avoid exploding the feature space through one-hot encoding.

13.4 Feature Scaling

The dataset includes both continuous engineered features (such as Duration_mins) and one-hot encoded binary features. To ensure fair learning for distance/margin-based algorithms like Linear Regression, KNN, and SVR, feature scaling was performed using `StandardScaler`.

13.5 Selecting the Best Model

Different machine learning algorithms produced different prediction results. To identify the most suitable model, six regression algorithms were trained and compared using MAE, MSE, RMSE, and R^2 Score.

13.6 Hyperparameter Tuning

The default model parameters may not provide the best performance. `GridSearchCV` was used to optimize the hyperparameters of the best-performing model and improve its prediction capability.

13.7 Outliers in Price

Some very high fare values were present in the data. These were inspected using boxplots and the IQR method but were retained, since they represent valid premium or long-haul fares rather than data entry errors.

13.8 Avoiding Overfitting

Tree-based models can perform very well on training data but poorly on unseen data. This challenge was addressed by evaluating all models on a separate test dataset and selecting the model with the best generalization performance.

13.9 Interpreting Feature Importance

Identifying the booking details that most influence fare was essential for business understanding. The Random Forest Feature Importance technique was used to determine the most significant features contributing to price prediction.

14. Final Conclusion

14.1 Project Objective

The primary objective of this project was to develop a machine learning model capable of predicting flight ticket fares based on booking details such as airline, route, duration, number of stops, and journey date.

14.2 Exploratory Data Analysis (EDA)

A comprehensive Exploratory Data Analysis (EDA) was conducted to understand the dataset by examining its structure, checking for missing values and duplicates, analysing the price distribution, identifying outliers, and exploring relationships between the features and the target variable.

14.3 Model Development and Evaluation

Six machine learning regression models — Linear Regression, Decision Tree, Random Forest, K-Nearest Neighbors (KNN), Support Vector Regressor (SVR), and Gradient Boosting — were trained, evaluated, and compared using MAE, MSE, RMSE, and R^2 Score to identify the best-performing model.

14.4 Model Optimization

The best-performing model was further optimized through Hyperparameter Tuning and is recommended for predicting the fares of new flight bookings due to its high predictive performance, improved accuracy, and reliability.

14.5 Feature Importance Analysis

Feature Importance analysis identified Total_Stops, Duration, Airline, and journey timing as the most influential factors affecting flight fare. These insights provide valuable information for both customers and travel businesses.

14.6 Machine Learning Approach and Insights

The project also demonstrates the importance of proper data preprocessing, feature engineering, model selection, and evaluation techniques in developing reliable machine learning regression systems. The comparative analysis of multiple algorithms helped in understanding the strengths and performance of different approaches for solving real-world business problems.

14.7 Business Impact and Conclusion

Overall, this project proves that machine learning can effectively support flight fare prediction, enabling customers and travel businesses to make data-driven booking and pricing decisions, reduce pricing errors, and improve planning around flight costs.

14.8 Future Enhancements

Future enhancements can include integrating larger and more recent flight fare datasets, applying advanced ensemble/boosting techniques (e.g. XGBoost, LightGBM), incorporating additional real-world features such as seasonal demand and fuel prices, and deploying the optimized model as a web application or API for real-time fare prediction.

15. Project Links

The following links provide access to the complete project resources, including the source code, dataset, trained models, presentation, and supporting documentation.

15.1 GitHub Repository

Access to the complete source code, Jupyter Notebook, README, and project files:

github.com/Sameekshagithub/PRCP-1025-Flight-Fare-Price-Prediction-Using-Machine-Learning-Regression.git

15.2 Google Drive

Access to the project presentation (PPT) and report:

drive.google.com/drive/folders/1Hp_r_xMfYt7KTZ2XN35BsMCoUasQRBP

15.3 Dataset Source

d3ilbtxij3aepc.cloudfront.net/projects/CDS-Capstone-Projects/flight-fare.zip

15.4 Acknowledgement

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